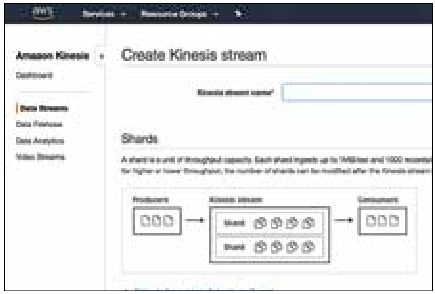
Now let us take a look at how to go about flowing data along your stream.

A Kinesis data stream usually has at least two components, a Data Producer and a Data Consumer, we will briefly touch upon these two and their uses in the following sections.



Start off your journey with Kinesis.

Data Producer

The data producer can be anything that adds data to Streams, such as an EC2 instance, client browser, or mobile device. In most real-life scenarios, the data consumer and data producers are two separate applications using Streams to communicate asynchronously.

To add data into a stream, producers call the PutRecord operation. Each PutRecord call requires the stream name, a partition key, and the data record that the producer is adding to the stream. The stream name determines the stream in which the record will reside. The partition key is used to locate the shard of the stream that the data will then be added to.

The choice of partition key depends on the application. The only thing to note is that the number of partition keys should be much greater than the number of shards. A high number of partition keys to shards helps the stream distribute data evenly across the shards in your stream.

Data Consumer

Data consumers are workers that retrieve and process data from the stream. Each consumer reads data from a particular shard. Consumers pull data from a shard using the GetShardIterator and GetRecords operations.

A shard iterator denotes the position of the stream and shard from which the each consumer will read. A consumer gets a shard iterator when it starts reading from a stream or changes the position from which it reads records from a stream. To get a shard iterator, consumers must provide a stream name, shard ID, and shard iterator type. The shard iterator type allows it to specify where in the stream it would like to start reading from (such as from the start of the stream where the data is arriving in real-time). The stream has an option to return the records in a batch, whose size you can control using the limit parameter.